

Model Card - FairCredit Jurisdiction Navigator Prototype

1. Model Overview

Item	Description
Model name	FairCredit Jurisdiction Navigator Prototype
Use case	AI-supported credit card application assessment
Target entity	American Express
Jurisdictions	United States and European Union
Model type	Logistic regression
Data	Synthetic credit application data
Output	Approved, Manual Review, or Declined
Purpose	Demonstrate jurisdiction-aware compliance logic

This is an academic prototype. It does not use real American Express customer data and does not represent American Express's actual credit policy.

2. Intended Use

The model is intended to show how an AI-supported credit decision can be interpreted differently across jurisdictions. Intended users: compliance teams; model risk teams; credit risk teams; governance reviewers; senior management.

The model supports demonstration of: synthetic credit scoring; US adverse action explanation; EU fairness and drift monitoring; human review triggers.

3. Data and Features

The prototype uses 2,000 synthetic credit application records. The test sample contains 600 scored applications.

Main model features:

Feature	Purpose
annual_income	Repayment capacity
debt_to_income	Debt burden
credit_history_years	Credit record depth
missed_payments_12m	Recent repayment issues
credit_utilization	Existing credit pressure
employment_status	Income stability
spending_volatility	Behavioural risk
requested_credit_limit	Requested exposure

The variable synthetic_group_proxy is excluded from training and used only for post-model fairness monitoring.

Target variable:

Target	Meaning
good_repayment_12m = 1	Simulated good repayment
good_repayment_12m = 0	Simulated poor repayment

4. Model and Decision Logic

I used logistic regression because it is simple, transparent, and suitable for a regulatory prototype. The aim is not maximum predictive accuracy, but clear demonstration of compliance logic.

Probability Range	Decision
≥ 0.62	Approved
0.48–0.62	Manual Review
< 0.48	Declined

Test sample result:

Decision	Count	Share
Approved	500	83.3%
Manual Review	50	8.3%
Declined	50	8.3%

The high approval rate reflects the synthetic data and simplified thresholds. It should not be treated as a real approval rate.

5. Jurisdiction-Aware Outputs

Mode	Compliance Logic	Output
US Mode	Individual explanation	Adverse action reasons
EU Mode	System governance	Fairness metrics, drift alerts, human review

US Mode:

For declined or borderline applicants, the tool generates reasons such as:

- high debt-to-income ratio;
- limited credit history;
- recent missed payments;
- high credit utilisation;
- requested credit limit high relative to income.

Key question: Can the firm explain this specific negative decision clearly and accurately?

EU Mode:

The tool monitors portfolio-level risks, including approval-rate disparity, error-rate differences, average model probability by group, and drift.

Fairness output:

Group	Records	Approval Rate	Avg. Model Probability
Group A	346	88.15%	0.841
Group B	169	73.96%	0.749
Group C	85	82.35%	0.814

The approval-rate disparity ratio is 0.839, above the 0.80 alert threshold. No hard fairness alert is triggered, but Group B remains a monitoring signal.

6. Drift Monitoring

Variable	PSI	Review Required
annual_income	0.032	No
debt_to_income	0.237	Yes

Variable	PSI	Review Required
credit_utilization	0.178	Monitor
missed_payments_12m	0.092	No

The strongest signal is debt-to-income, with PSI of 0.237, above the 0.20 review threshold. This would trigger human model governance review in EU mode.

7. Human Review Triggers

Human review is required when:

Trigger	Reason
Manual Review decision	Applicant is close to threshold
Declined decision in US mode	Explanation must be checked
Fairness threshold breach	Possible group-level disparity
Material PSI drift	Population has shifted
Jurisdictional uncertainty	Legal or compliance judgement needed

The model supports human judgement. It does not replace it.

8. Limitations

This prototype:

- uses synthetic data only;
- does not represent real American Express customers;
- uses simplified thresholds;
- uses synthetic group proxies;
- does not provide legal advice;
- does not make final credit decisions;
- is not production-validated;
- does not cover all US or EU requirements.

9. Governance Plan

If developed further, the tool should include:

Area	Action
Performance	Track AUC, accuracy, false positives, false negatives
Fairness	Monitor group approval and error-rate differences
Drift	Track PSI for key variables
Explanations	Review adverse action reasons
Rules	Version jurisdictional logic
Audit	Store reviewer notes and decision history

Final Assessment

The model is suitable for demonstrating the core design of FairCredit Jurisdiction Navigator. It shows how a credit decision can be translated into US adverse action explanations and EU system-level governance checks.

The model is not production-ready. Its value is to demonstrate how a jurisdiction-aware RegTech tool can make regulatory differences visible in the credit decisioning workflow.